# MAMDOUH ATTIA

Software Engineer

mamdouhattia31@gmail.com | +201275400234 | linkedin.com/in/mamdouh-atia | github.com/Mamdouh-Attia | Egypt | Military service : Completed

## PROFESSIONAL SUMMARY

**Computer Engineering** graduate (**Cairo** University) with **3+ years** building **cross-platform** mobile apps for real clients, systems, and AI-powered software. Delivered end-to-end solutions using **Flutter**, **Python**, and **C/C++.** Strong foundation in systems programming, **data structures**, **algorithms**, and **ML**. **200+** hours technical instruction across **ML**, **Python**, and **Flutter** programs.

## EDUCATION

**Cairo University — Faculty of Engineering, Computer Engineering** *2019 – 2024*

*Grade: Very Good with Honors | Graduation Project Grade: Excellent (A+)*

## PROFESSIONAL EXPERIENCE

**Flutter Developer — Freelance, PrimaLab, Itqan Community** *Sep 2023 – Present*

- Delivered +3 client Flutter apps (HyperMate, VolunTeen, Check-It) integrating Hardware IoT sensors (ESP8266), AI chatbots (GPT-3), Firebase, REST APIs, QR/barcode scanning.
- Contributed to open-source mushaf-imad-flutter package (PR #27); Integrating Quran.com's audio API : OAuth 2.0, hybrid audio engine, Stream Transformers, 80+ Mocktail tests, 86% latency reduction (by token caching).
- Applied MVC, MVVM, Clean Architecture; owned full SDLC from Figma wireframes to tested releases.

**Technical Instructor — DEPI/MCIT @ Harvest, Maspero Academy, PrimaLab** *Aug 2023 – Jun 2025*

- **DEPI Initiative**: Led 100+ hrs ML Engineering track for 15+ engineers (statistics, scikit-learn, evaluation, Azure AI); designed and graded 17 tasks and final capstone.
- **Maspero Academy**: Delivered 20+ hrs Python basics for 35 students. **PrimaLab**: 50+ hrs Dart/Flutter across 28 sessions for 10+ students.

**Flutter Developer Intern — Orange Digital Center Software House** *Jun 2023 – Sep 2023*

- Worked in Agile squads on 2 internal mobile apps and 1 hackathon e-wallet. Translated Figma designs to Flutter UI using MVVM + Bloc/Cubit.

**IT Support Specialist — Future Of Egypt (Mandatory Military Service)** *Dec 2024 – Nov 2025*

- Technical support for hardware, software, and networks in a government operational environment.

## SELECTED PROJECTS

**Mushaf-IMAD Package | Dart, REST API, OAuth 2.0, Streams, Mocktail** *Feb – Mar 2026*

- 86% latency reduction via Single-Trip optimization; hybrid timing hierarchy (RAM → Local → API) for zero-latency audio playback sync.
- +70 atomic commits of 5000+ lines, 80+ async Mocktail tests; published architectural guides for open-source maintainability.

**Predicto (Graduation Project, A+) | Python, PyTorch, OpenCV, Desktop GUI** *Aug 2023 – Jul 2024*

- Team lead: built vPredicto open-source library for Video Frame Prediction with full PyQt desktop app (model selection, video input, output).
- **Implemented from scratch:** *SimVP* [co-implemented] (CNN Encoder-Translator-Decoder, TPU-optimized, 5× training speedup via checkpointing) and PredNet (ConvLSTM predictive coding, resolved critical dimension-mismatch bugs).

**VolunTeen | Flutter, Firebase, GPT-3, Provider, MVC** *Sep 2023 – Jan 2024*

- Solo developer for real client: 30–40 responsive screens, GPT-3 chatbot, Firebase storage, in-app video player, WhatsApp/URL launcher.

**HyperMate | Flutter, C, ESP8266, Wi-Fi, GPT-3** *Nov – Dec 2023*

- IoT stress-monitoring app with real-time GSR sensor readings over Wi-Fi, personalized exercise, AI chatbot. Also authored Arduino firmware.

**Distributed File System (Go-DFS) | Go, gRPC, Protocol Buffers, TCP** *Mar – Apr 2024*

- Master-Worker architecture: gRPC heartbeat monitoring, auto-replication engine (factor 3), Data Keeper self-scanning, and MP4 upload/download pipelines via raw TCP.

**OS Scheduler | C, Linux, IPC, Memory Management** *Apr 2022*

- Led Process Generator: fork()/exec() lifecycle, IPC Message Queues. Designed Buddy Allocator (1024-byte pool, recursive splitting/merging, allocate() + deallocate() functions).

**Gamingo Engine + Drive Me Home Game | C++, OpenGL, GLSL, OOP** *Mar – May 2023*

- Custom 3D game engine: Role: Shader, Texture, Material, Sampler classes + GLSL programmable shading. Collision detection, game-state machine, post-processing shaders (Sepia, CRT, Inversion).

**MIPS Processor | Verilog, Computer Architecture, FPGA Simulation, Modelsim** *Dec 2022*

- 5-stage pipelined MIPS processor. Role: decode stage, top-level integration (5-stage buffers), control unit ISA mapping, hazard detection and forwarding verification.

## SKILLS

**Programming:** Dart · C/C++ · Python · Java · Go (basic) | **Web (basic) :** HTML · CSS · JS

**Mobile / Frontend:** Flutter · Provider · Bloc/Cubit · MVC · MVVM · Clean Architecture · Streams · BAAS (Firebase - Supabase) · REST APIs

**AI / ML:** PyTorch · TensorFlow · scikit-learn · OpenCV · Computer Vision · NLP · pandas · numpy · Machine Intelligence · ML · DL

**Systems:** Linux · Git/GitHub · GitHub Actions · Algorithms · Data Structures · Design Patterns · OOP · SOLID SDLC · Agile/SCRUM · gRPC

**Hardware:** ESP8266 · Wi-Fi · Arduino Firmware · Hardware Sensors · Embedded Systems · IC Design · Verilog · VLSI

## CERTIFICATES

GitHub Professional Certificate (2026)

Flutter Internship — ODC (2023) ·

Machine Learning Specialization — Coursera (2024)

HCIA-AI — Huawei International (2021)